A BHATIA–ŠEMRL OPERATOR WITH INFINITE-DIMENSIONAL NORMING SPAN

AGENT LANE 12

Source and target

Jayanarayanan–Rajpopat prove that if a nonzero operator T has the Bhatia–Šemrl property and $\operatorname{span}(M_T)$ is finite dimensional, then $\|T\|_e < \|T\|$. In Remark 3.3 of [1], they ask whether the finite-dimensionality hypothesis might follow automatically from the Bhatia–Šemrl property.

In the next two results, we give a sufficient condition under which the Bhatia-Šemrl property is related to the essential norm of an operator.

Proposition 3.2. *Let X and Y be Banach spaces. Let $T \in B(X,Y)$ be a nonzero operator such that T satisfies the Bhatia-Šemrl property. If $\operatorname{span}(M_T)$ is finite dimensional, then $\|T\|_e < \|T\|$.*

Proof. Assume that $\|T\|_e = \|T\|$. Then $T \perp_B K(X,Y)$. Let $X_0 = \operatorname{span}(M_T)$. Since X_0 is a finite dimensional subspace of X, it is complemented in X. That is, there exists a projection $P \in B(X)$ such that $R(P) = X_0$. Hence P is a finite rank projection. Now consider the operator $K : X \to Y$ defined as $K = TP$. Clearly, $K \in K(X,Y)$ and hence $T \perp_B K$. But for any $x \in M_T$, we have $Kx = TPx = Tx$. Hence, there is no $x \in M_T$ such that $Tx \perp_B Kx$, which contradicts the hypothesis. □

Remark 3.3. In view of the Proposition 3.2, it is natural to look at the class of operators T on Banach spaces for which $\operatorname{span}(M_T)$ is finite dimensional. In particular, one can ask whether every operator satisfying the Bhatia-Šemrl property necessarily has $\operatorname{span}(M_T)$ finite dimensional. In the Hilbert space setting, Theorem 1.2 shows that such operators indeed have $\operatorname{span}(M_T)$ finite dimensional.

Now let us recall the definition of the compact approximation property.

Here

$$M_T = \{x \in S_X : \|Tx\| = \|T\|\},$$

and T has the Bhatia–Šemrl property when, for every operator A with $T \perp_B A$, there exists $x \in M_T$ such that $Tx \perp_B Ax$.

Result

The answer to the question in Remark 3.3 is negative, even for a scalar-valued operator on a simple reflexive Banach space. Take

$$X = \mathbb{R} \oplus_\infty \ell_2, \qquad f : X \longrightarrow \mathbb{R}, \qquad f(a,h) = a.$$

Then f has the Bhatia–Šemrl property, but $\operatorname{span}(M_f) = X$ is infinite dimensional.

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Proof intuition

The unit ball of $\mathbb{R} \oplus_\infty \ell_2$ has two large flat faces on which the first-coordinate functional attains its norm. Their linear span contains the entire ℓ_2 direction. At the same time, the dual norm is an ℓ_1-sum norm. Thus orthogonality of $f = (1, 0)$ to an arbitrary $g = (\alpha, k)$ is the elementary inequality $|\alpha| \leq \|k\|_2$. This inequality lets us choose a point on the norming face at which g vanishes, which is exactly the pointwise witness required by the Bhatia–Šemrl property.

Theorem 1. *Let $X = \mathbb{R} \oplus_\infty \ell_2$ and let $f \in X'$ be given by $f(a, h) = a$. Then f, regarded as an operator from X to $\mathbb{R}$, has the Bhatia–Šemrl property, while*

$$\operatorname{span}(M_f) = X.$$

In particular, the span of M_f is infinite dimensional.

Proof. The norm on X is

$$\|(a, h)\| = \max\{|a|, \|h\|_2\},$$

so $\|f\| = 1$ and

$$M_f = \{(\varepsilon, h) : \varepsilon \in \{-1, 1\},\ \|h\|_2 \leq 1\}. \tag{1}$$

The vector $(1, 0)$ lies in M_f. If $\|h\|_2 \leq 1$, then both $(1, h)$ and $(1, 0)$ lie in M_f, and

$$(0, h) = (1, h) - (1, 0).$$

After scalar multiplication this gives $\{0\} \oplus \ell_2 \subseteq \operatorname{span}(M_f)$, while $(1, 0)$ supplies the first coordinate. Hence $\operatorname{span}(M_f) = X$.

It remains to prove the Bhatia–Šemrl property. Identify

$$X' = \mathbb{R} \oplus_1 \ell_2.$$

Let $g = (\alpha, k) \in X'$ and suppose that $f \perp_B g$. For every $\lambda \in \mathbb{R}$,

$$\|f + \lambda g\| = |1 + \lambda\alpha| + |\lambda|\, \|k\|_2 \geq 1. \tag{2}$$

We claim that

$$|\alpha| \leq \|k\|_2. \tag{3}$$

Indeed, if $\alpha > \|k\|_2$, then for all sufficiently small $t > 0$,

$$\|f - tg\| = 1 - t\alpha + t\|k\|_2 < 1,$$

contradicting (2). If $\alpha < -\|k\|_2$, the same contradiction follows from $\|f + tg\| < 1$ for small $t > 0$. This proves (3).

Suppose first that $k \neq 0$, and set

$$h = -\frac{\alpha}{\|k\|_2^2} k.$$

By (3), $\|h\|_2 \leq 1$. Thus $x = (1, h)$ belongs to M_f by (1), and

$$g(x) = \alpha + \langle k, h \rangle = \alpha - \alpha = 0.$$

If $k = 0$, then (3) gives $\alpha = 0$, and we take $x = (1, 0) \in M_f$. In either case, $f(x) = 1$ and $g(x) = 0$. Since a nonzero real scalar is Birkhoff–James orthogonal precisely to the zero scalar, we have $f(x) \perp_B g(x)$. This is exactly the Bhatia–Šemrl property for f. □

Relation to known results

Kim–Lee proved that every functional on a Banach space has the Bhatia–Šemrl property if and only if the space is reflexive [2]. Since $\mathbb{R} \oplus_\infty \ell_2$ is reflexive, that theorem independently confirms the property used above. The calculation (2) gives a direct, self-contained proof for this example and isolates the flat norming face that answers Remark 3.3.

The example does *not* answer the source's broader Question 1.9, which asks whether the Bhatia–Šemrl property always implies $\|T\|_e < \|T\|$. The range of f is one dimensional, so f is compact and $\|f\|_e = 0 < 1 = \|f\|$.

Verification and novelty bounds

The argument uses only the elementary duality $(\mathbb{R} \oplus_\infty \ell_2)' = \mathbb{R} \oplus_1 \ell_2$. The crucial implication $f \perp_B g \Rightarrow |\alpha| \le \|k\|_2$ was checked in both scalar directions, and the resulting witness h has norm at most one exactly when that inequality holds.

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2512.15208 and the phrases "Bhatia–Šemrl", "span(M_T)", and "finite dimensional norm attainment span". No existing packet for this question was found. Bounded web searches through August 9, 2026, found the source paper and the Kim–Lee reflexive-space theorem, but no publication explicitly answering Remark 3.3. Because the counterexample is a short consequence of a theorem already cited by the source, its mathematical validity is high-confidence while its novelty should be labeled medium-low pending expert review.

References

[1] C. R. Jayanarayanan and R. R. Rajpopat, *On Bhatia–Šemrl property, strong subdifferentiability and essential norm of operators on Banach spaces*, arXiv:2512.15208, 2025.

[2] S. K. Kim and H. J. Lee, *The Birkhoff–James orthogonality of operators on infinite dimensional Banach spaces*, Linear Algebra Appl. 582 (2019), 440–451.